CHAPTER

# 8

# Advancing Early Diagnosis of Parkinson's Disease: The Role of Deep Learning and Innovative Diagnostic Tools

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## 1. Introduction

Parkinson's Disease (PD) is a progressive neurodegenerative disorder characterized by motor dysfunctions such as bradykinesia (slow gait movement), resting tremors, muscle rigidity, and postural instability. Diagnosing Parkinson's disease (PD) continues to pose a significant clinical challenge, primarily due to the lack of definitive biological markers and the reliance on subjective clinical assessments. In most cases, diagnosis is based on the identification of cardinal motor symptoms, which generally emerge only after substantial neurological deterioration has taken place—by which point approximately 70% of neurons have already degenerated. (Suchowersky *et al.*, 2006). Moreover, clinicians must often rely on exclusion criteria and observable symptoms to differentiate PD from other disorders, leading to a substantial risk of misdiagnosis. The use of standardized clinical tools such as the Unified Parkinson's Disease Rating Scale (UPDRS) is limited by its dependence on the examiner's interpretation and the patient's subjective feedback. Despite ongoing research, existing methodologies largely fail to detect PD at an early stage, where therapeutic interventions could be most effective. There is a critical gap in the availability of objective, quantifiable diagnostic tools that can support early, accurate identification of the disease. Furthermore, conventional approaches are inadequate in distinguishing PD from other conditions with overlapping motor symptoms, particularly in the prodromal or mild stages. To address these limitations, this chapter explores the integration of Artificial Intelligence (AI) and sensor-based technologies—specifically, handwriting analysis and Inertial Measurement Unit (IMU)-based wearable systems for early detection of PD. The novelty of this approach lies in its dual-modality framework: AI-driven handwriting analysis captures subtle motor impairments such as micrographia, while IMU-based wearables quantify gait dynamics, tremor, and movement asymmetry in real time. Together, these tools offer a non-invasive, data-driven pathway to support clinicians in identifying early signs of PD. PD is a challenge to diagnose, as there are no well-established biological markers to determine the presence of the disease. Various symptoms of PD is shown in Figure 8.1 (Breen, Michell and Barker, 2011). There is no definitive diagnosis of PD, so the diagnosis is clinical and based on slow gait movement (bradykinesia).

## 2. Related works

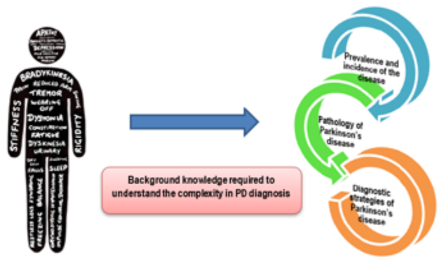


**Fig. 8.1 Symptoms of Parkinson's Disease** Symptoms of Parkinson's Disease No Permission Required

Emerging technologies are transforming the diagnosis of Parkinson's disease by enabling earlier, more accurate, and less invasive detection. Traditional methods often identify the disease only after significant brain damage has occurred. Innovations in imaging, biomarkers, and wearable devices now offer promising diagnostic tools. These advances pave the way for timely interventions and improved patient outcomes.

*a) Handwriting analysis for PD*

Research studies for the early detection of Parkinson's disease have seen significant advancements owing to the recent developments in Artificial Intelligence. Several studies have been carried out to identify PD symptoms through non-invasive modalities such as speech, handwriting etc. This section highlights some prominent works carried out for the detection of Parkinson's symptoms using AI. Mercaldo et.al.(2024), proposed a method employing ResNet-50 and DenseNet for PD detection. They employed a dataset of 3,991 images related to spiral drawing tests, with 1,995 images representing patients diagnosed with PD and the remaining 1,996 images representing individuals without the disease condition, to differentiate between PD patients and healthy individuals (Mercaldo et al., 2024). DenseNet achieved better performance compared to ResNet-50. Bernardo, et al. used SqueezeNet model as a feature extractor, and Support Vector Machine (SVM) as a classifier to distinguish individuals affected by PD from healthy individuals (Bernardo *et al.*, 2021). The dataset used for training and testing consists of 514 handwritten drawings of Archimedes' spiral images derived from heterogeneous sources (digital and paper-based), from which 296 correspond to PD patients and 218 to healthy subjects. Aversano et.al. (2022), proposed a method for early detection of PD using a combination of an Echo State Network (ESN) and a Multi-Layer Perceptron (MLP) (Aversano *et al.*, 2022). The study utilized the UCI Parkinson Disease Spiral Drawing Dataset, comprising 77 patients (62 PD, 15 healthy) tested on both static and dynamic spiral tests. Another study by Chandra et.al.(2021), analyzed the Parkinson's Handwriting dataset, which included static and dynamic spiral drawings from 62 PD patients and 15 healthy controls(Chandra *et al.*, 2021). The authors utilized a Random Forest (RF) classifier with features extracted from the geometric and kinematic properties of the spiral drawings.

*b) Tremor analysis using Imaging Techniques*

Various neuroimaging techniques are used to figure out the dopamine deficiency in central nervous symptoms. The neural structure is imaged using a combination of neural imaging tools for the early diagnosis of PD, like Positron Emission Tomography scan (PET), Transcranial Sonography (TCS), Single Photon Emission Computed Tomography (SPECT), and Magnetic resonance imaging (MRI). The technique is used as a tool to study the dopamine level in the neurons, as shown in Figure 8.2. This study investigates an approximate range of dopamine levels that may either increase or decrease. But this method lacks the ability to investigate the effects of this on motor and non-motor symptoms.

The dopamine deficiency in the Central Nervous System, (CNS) will affect the communication of control signals and sensory information. This barrier of data transmission will result in the loss of control over a variety of organs in the body. In some cases, control signals will not be sent, and in some cases, feedback signals will not be received. This miscommunication created between the CNS, Peripheral Nervous System (PNS), and organs will lead to Parkinson's disease with various motor and non-motor symptoms. These symptoms are common

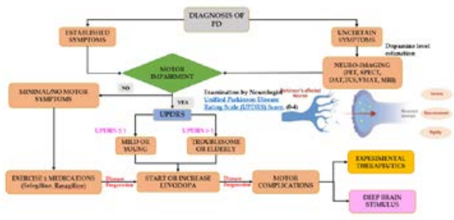


**Fig. 8.2 Strategies for Diagnosing Parkinson's Disease**
Strategies for Diagnosing Parkinson's Disease No Permission Required

in other neurodegenerative diseases and make the diagnosis of PD challenging. Over the past decade, clinical diagnostic accuracy for PD has surpassed 90%, largely due to consistent long-term monitoring. However, early-stage detection remains crucial for mitigating disease progression. In response, researchers are actively exploring innovative diagnostic strategies aimed at enhancing early diagnostic precision. The rapid advancements in AI and Machine Learning (ML) have played a pivotal role in biomedical data analysis, demonstrating promising results across various applications, such as the early detection of Alzheimer's disease using MRI. In particular, the integration of neuroimaging techniques with machine learning and deep learning algorithms has significantly improved the automated detection and classification of PD.

Pablo Martinez-Martin et al (2018), have validated the simple screening tool of a questionnaire for early diagnosis of Advanced Parkinson's Disease (APD) by observing the daily activity(Martinez-Martin *et al.*, 2018). Prashanth et al (2017), have developed an effective system with the features resulting from the shape and surface fitting process for improving the classification accuracy of early PD. The averaged SPECT image slice is involved in extracting the features. Skills-Based Routing (SBR) values are also incorporated along with these features. Twenty-five shape and surface fitting features have been identified by the feature importance scores from the Random Forest technique as having a greater discriminative power than SBR. The techniques adopted for classification include SVM, Boosted Trees, Random Forest, and Naive Bayes on the reduced dataset, with SVM achieving the highest accuracy of 97.29%. (Prashanth *et al.*, 2017). Warwick R. Adams (2017), has developed a system that diagnoses PD in the early stage. Hand and finger movement characteristics are taken from 32 mild PD patients and 71 healthy controls. Since keystroke timing information does not correlate with motor symptoms of PD, the system could not discriminate the PD from the related disease, SWEDD(Adams, 2017). Hui-Ling Chen et al (2016), have explored the potential of the Kernel Extreme Learning Machine (KELM) using the key parameters for diagnosing the early stage of PD. The KELM parameters 'C' and 'γ' are investigated empirically with the help of the University of California, Irvine (UCI) database (Chen *et al.*, 2016).

Mehrbakhsh Nilashi et al. (2016) developed a model that gained significant attention as it aims to predict Parkinson's disease even in its early stages and track its progression using noise removal, clustering, and predictive analysis. The input data are drawn from the UCI machine learning repository, including UPDRS scores. The clustering technique, Expectation Maximization (EM), and the dimensionality reduction technique, Principal Component Analysis (PCA) are used for noise removal. The Adaptive Neuro-Fuzzy Inference System (ANFIS), as well as Support Vector Regression (SVR), are used to predict the disease in its initial stage. The combined application of EM, PCA, ANFIS, and SVR methods has been validated to produce results that significantly facilitate the diagnosis of Parkinson's disease. Although it is effectively implemented as a clinical decision support system for PD treatment and progression, the system introduces complexity in its design(Nilashi, Ibrahim and Ahani, 2016). The model developed by Prashanth et al (2014), aims to assist in classifying and predicting PD in its early stage. The SBR is computed from SPECT images which are drawn from the Parkinson's Progression Markers Initiative (PPMI) database. This is used to build a prediction/ prognostic model for the early stage of PD. The results of predictive models had the potential to discriminate early PD from Healthy Control (HC) automatically using RBF-ELM with an accuracy of 90.8%. However, these predictive models are subject to SBR values of the striatum(Prashanth *et al.*, 2014). Antonio Augimeri et al (2016), have proposed a fully automated technique for the analysis of three-dimensional DaTscan images. The scan images of 31 PD & 12 healthy subjects are selected and preprocessed to acquire the required features from it. The intensity, shape, and symmetry features are estimated from the striatum. The extracted multimodal features have the greater discriminating power when the SVM classifier is used (Augimeri *et al.*, 2016). Saurabh Jain et al (2014), have proposed a system to enhance the clinical utility in the diagnosis of PD and tracking its progression using quantification-based pattern recognition. The surface registration algorithm, which belongs to Large Deformation Diffeomorphic Metric Mapping (LDDMM), is implemented to extract 3D Shape/texture metrics. The common coordinate systems are obtained based on MR images to estimate population-based templates for each structure. The high variance of intensities is estimated across the subjects using Principle Component Analysis (PCA) (Jain *et al.*, 2016).

Antonio-Rubio et al (2015), conducted a study and found that a non-invasive cold stress test was a reliable and straightforward method for detecting sympathetic cutaneous dysfunction in Parkinson's disease patients. This study included 15 people with Parkinson's disease and 20 healthy people. The impact of Parkinson's disease on SVR was studied using a variety of PD features. The experiment was done with both hands, one immersed in cold water (IH) and the other not (NIH), regardless of motor laterality. When compared to controls, PD individuals had a lower mean baseline hand temperature ($p = 0.037$), and they did not exhibit a regular cooling pattern(Antonio-Rubio *et al.*, 2015). Yuichi Mitsui et. al (2020), introduced an image-processing technique to assess the tremor features of PD subjects. The finger-to-nose touch exercise is done to find the motor features like amplitude and frequency. The clinical assessment of cerebellar disorder, Essential Tremor (ET), and healthy subjects was done. The extracted assessment data was classified using machine learning models like Linear Discrimination, Logistic regression, support vector machine, and K-Nearest neighbor algorithm (Mitsui *et al.*, 2020). Purup. M. et al (2020), Peripheral autonomic dysfunction was assessed in 20 Parkinson's disease (PD) patients and compared to 19 healthy controls to identify γ-synuclein deposits. During the Cold Stress Test (CST), the temperature of the hands, arms, and legs was measured. Bhavana Palakurthi et al (2019), contributed to the diagnosis of postural instability by imaging White Matter Hyperintensities (WMHs), Gray Matter (GM) atrophy, and basal ganglion abnormalities. The author also discussed some of the reported factors for an early PI diagnosis, including age, nervous system lesions, genetic mutations, abnormal proprioception, impaired reflexes, and altered biomechanics(Palakurthi and Burugupally, 2019a) (Palakurthi and Burugupally, 2019b). Yuichiro Omura et al. (2023) proposed a neural controller model that integrates two types of control mechanisms: feedforward control (representing muscle tone as a vector) and feedback control using proprioceptive and vestibular sensory information. The results of the computational model suggested that patients with PD with increased muscle tone display less sway in abnormal postures than in other postures. The findings indicate that patients with Parkinson's disease must maintain an abnormal posture once their muscle tone reaches that of an anterior leaning posture, as their swing is greater in other postures than in the abnormal posture(Omura *et al.*, 2023).

## 3. Proposed System

During the clinical assessment of the PD subjects, various baseline and PD-specific characteristics were included. These features define an individual's fitness, PD symptoms, and other medical complications. The clinical evaluation of PD subjects is done to understand the tremors in the initial stages of the clinical onset of PD. The following categories of data are considered in this study:

1. Cohort data analysis: Baseline characteristics like age, gender, BMI, Hypertension, Diabetes, etc.
2. History of Medical complications: The information about disease diagnosis time, PD assessment, and family history with PD or other neurological disorders.
3. Motor severity assessment: Movement Disorders Society - Unified Parkinson's Disease Rating Scale (MDS-UPDRS) severity score for various motor functionality assessments.
4. Non-motor assessment: MDS-UPDRS score for non-motor symptoms like sleep disturbance, fatigue, swallowing, autonomic dysfunction, and mood disorders.

**Table 8.1** Baseline characteristics of 3 different groups of neurological disorders

| | PD Subjects (n=5) | Essential tremor subjects (n=5) | Physiological tremor subjects (n=5) |
|---|---|---|---|
| Age (years): mean ± SD (range) | 45.8 ± 6.35 | 52.2 ± 6.76 | 57.3 ± 14.5 |
| Sex (male/female) | 3/2 | 2/3 | 4/1 |
| BMI (kg/m2): mean ± SD | 26.6 ± 11.77 | 25.46 ± 11.21 | 23.54 ± 11.43 |
| Diabetes mellitus (yes/no) | 1/4 | 2/3 | 3/2 |

*No Permission Required*

All these features taken during the assessment represent the characteristics of PD, and it is obtained by the detailed questionnaire given in MDS-UPDRS. Among the motor and non-motor symptoms, the resting tremor is assessed using an image processing technique, and the handwriting assessment using micrographia. PD is often associated with tremors, rigidity, and bradykinesia (slowness of movement), which can significantly affect a person's handwriting. The use of AI-based handwriting analysis for detecting Parkinson's disease relies on advanced techniques that help identify early signs of the disease by analyzing specific features of a person's handwriting. These techniques allow for the tracking of the disease's progression over time. Various approaches, including image processing, machine learning, time-series analysis, and signal processing, are employed to identify characteristics of handwriting that may be indicative of Parkinson's.

### 3.1 Data Acquisition and Subject Selection

To estimate the severity score of resting tremors in 15 neurological disorder subjects, the subjects were continuously monitored for 10 days with a vision-based sensor technique. The study included 15 subjects, and various baseline and PD characteristics are considered for the investigation as shown in the Table 8.1. By considering all these characteristics with the resting tremor frequency, the tremor UPDRS score was calculated. The estimation of the resting tremor UPDRS score was done using the traditional diagnostic technique. The results of the severity score were compared with the severity score obtained by measuring tremor frequency. The tremors of three different neurological disorders PD, ET and Physiological tremor (PT) are investigated in this study.

Image processing and feature extraction play a key role in the early detection and monitoring of Parkinson's disease. By analyzing handwriting samples, image processing tools extract quantitative characteristics that reveal trends associated with the disease. Stroke features, such as stroke length, speed, pressure, and direction, are examined to detect abnormalities linked to PD. Handwriting irregularities, such as tremors or micrographia (small, cramped handwriting), are closely analyzed. Factors like the shape and consistency of letters, as well as the speed and fluidity of writing, can provide important diagnostic clues. Parkinson's patients tend to write more slowly and with reduced fluidity, so measuring the time taken to form each character is a distinguishing factor. Variations in pen pressure can also be detected, revealing motor issues common in individuals with Parkinson's. ML algorithms are widely used in AI-based handwriting analysis to classify handwriting features and detect patterns associated with Parkinson's disease. Supervised learning techniques, such as decision trees, SVM, and random forests, use labeled handwriting data (healthy vs. Parkinson's) to train models that can recognize distinguishing patterns. More advanced approaches, like deep learning, utilize Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for analyzing sequential data (e.g., writing strokes) or spatial data (e.g., letter shapes). By selecting relevant features of handwriting—such as stroke velocity, pressure, and letter consistency—the accuracy of machine learning models can be significantly improved, providing a more reliable diagnosis. Time-series analysis is another critical technique used for tremor detection in handwriting. Tremors, which can vary in frequency (high or low), can be detected through the analysis of time-series data that tracks the pen's movement during writing. Dynamic Time Warping (DTW) is an algorithm used to align the time-series data of a patient's handwriting to a reference writing pattern. DTW is particularly useful for

**Fig. 8.3 A typical handwriting digitiser device** A typical handwriting digitiser device No Permission Required

comparing time-dependent handwriting features across different individuals, which aids in identifying subtle tremors or irregularities in movement. Signal Processing Techniques, such as the Fourier and Wavelet Transforms, are applied to detect irregularities in handwriting motion. Fourier transforms help analyze the frequency components of handwriting motion, allowing for the identification of tremors or abnormalities in movement frequency. Wavelet transforms, being a more advanced technique, can examine handwriting on multiple scales and periods, offering a deeper insight into the variations in movement associated with Parkinson's disease.

**3.1. Experimental setup**

The experimental setup is designed to facilitate the acquisition and analysis of multimodal data, specifically handwriting dynamics and resting tremor signals for the early detection and classification of Parkinson's Disease (PD). Given the lack of objective clinical tools for quantifying early motor symptoms such as micrographia and resting tremor, this setup integrates both wearable sensor technologies and digital input devices to collect precise, real-time physiological and behavioral data from study participants. To ensure comprehensive data capture, the experimental design incorporates two main subsystems: (1) a handwriting acquisition module using a digital tablet with pressure-sensitive stylus to record fine motor control during structured writing tasks (e.g., spiral drawing, signature replication), and (2) a tremor monitoring system using an Inertial Measurement Unit (IMU) and optional thermal or visible-light cameras to detect and quantify involuntary hand movements in the resting state. Both PD subjects and age-matched control participants are included to facilitate comparative analysis.

*a) Handwriting analysis using Micrographia*

Micrographia serves as a discriminative tool in the assessment of PD, offering valuable insights into the severity of motor impairments. Handwriting and drawing-based assessments, such as spiral tracing tasks, are frequently employed to capture physiological parameters like tremor amplitude, bradykinesia, and dyskinesia. These parameters facilitate differentiation between PD patients and healthy control groups, aiding in the early diagnosis and monitoring of disease progression. Pen pressure, speed, and other kinematic features extracted from handwriting samples have been shown to provide crucial biomarkers for identifying PD-related motor impairments. An advanced handwriting digitization system was developed to capture, analyze, and visualize handwriting data, enabling researchers to extract these features with high precision and reliability.

The relationship between handwriting and spiral drawing in the diagnosis of PD has been well-established in earlier studies (Kotsavasiloglou *et al.*, 2017). Spiral drawing kinematics provide key physiological parameters, such as tremor amplitude and the extent of bradykinesia and dyskinesia, aiding in PD assessment . These parameters enable differentiation between distal and proximal tremors and help distinguish individuals with PD from healthy control groups. Assessing the severity of these symptoms is crucial for clinical decision-making(Zham *et al.*, 2017). Pen pressure is another critical factor associated with handwriting analysis, offering valuable insights into distinguishing between individuals with PD and control groups (CG). By systematically analyzing handwriting and drawing tasks, researchers can leverage micrographia as a non-invasive diagnostic biomarker for PD. The handwriting digitizer system incorporates a commercially available Wacom Intuos Pro Large tablet coupled with a proprietary software application, as shown in Figure 8.3. This system records various handwriting parameters, including

- *pen up-down events,*

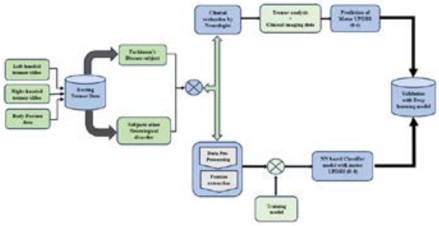

**Fig. 8.4 Steps involved in diagnosis of PD based on Imaging technologies** Steps involved in diagnosis of PD based on Imaging technologies No Permission Required


**Fig. 8.5 Experimental setup with Image sensors used for the study** Experimental setup with Image sensors used for the study No Permission Required

- *elapsed time,*
- *weighted average speed, and*
- *pen pressure.*

*b) Methods for tremor analysis using Imaging Technologies*

A detailed explanation of the process involved in investigating the use of the passive-vision-based technique for evaluating the resting tremor frequency of PD patients is shown in Figure 8.4. In this study, the mechanical hand arm vibrator was used on the healthy subjects to create mechanical vibration in the hands with a frequency range (2-15) hz similar to resting tremor. This tremor vibration was measured using Vision-based sensors. The image sensors are used to measure the vibration using image processing techniques. After manipulating the vibration frequency, each diagnostic tool's error rate was calculated by comparing the value with the input frequency, and the appropriate tool to calculate the resting tremor frequency was identified.

The proposed work is organized as mentioned below.

· A CMOS image sensor and an IR sensor were identified and instrumented for collecting the video samples of a hand-resting tremor.

· The image processing algorithm was developed to quantify the resting tremor frequency.

· Video Samples with the wearable Hand-arm vibrator were collected on healthy subjects, and the resting tremor frequency was calculated.

The block diagram for measuring the frequency of vibration in healthy subjects with HAV using image processing techniques is shown in Figure 8.5. The vibration of the hand is recorded as video in the GoPro 7 camera, which uses a CMOS image sensor to capture the video and a thermal image camera. The GoPro 7 motion camera supports a wide range of 4k resolution with 3840x2160 Pixels (2160p), 1920x1080 Pixels (1080p HD), and 1280x720 Pixels (720p HD). The sensor was suitable for collecting the mild rhythmic shaking movement of the subjects during HAV vibration due to the video stabilization feature provided by this camera.

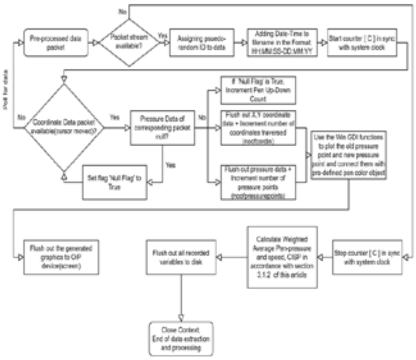


**Fig. 8.6 workflow for feature extraction in handwriting data** workflow for feature extraction in handwriting data
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### 3.2. Data analysis

To enhance the accuracy and robustness of Parkinson's disease (PD) detection, handwriting and tremor features can be combined using feature-level fusion or decision-level fusion strategies. In feature-level fusion, extracted features from handwriting tasks (such as velocity, pressure, stroke smoothness, and micrographia patterns) and tremor signals (such as amplitude, frequency, and power spectrum from IMU or thermal sensors) are concatenated into a single, unified feature vector. To process this multimodal data, advanced models such as ensemble methods (e.g., Random Forest, XGBoost) or multi-branch neural networks can be employed. These architectures are capable of learning shared and task-specific representations from each modality, improving model generalizability and resilience to missing or noisy data. The deep learning framework allows for parallel processing of handwriting and tremor features before merging them in higher layers for integrated decision-making. Finally, correlation analysis is conducted between the model predictions and clinical UPDRS scores to evaluate the clinical relevance and interpretability of the system. High correlation values support the validity of the AI-based predictions, ensuring the system is not only technically effective but also clinically meaningful. This approach bridges the gap between subjective clinical evaluations and objective, technology-driven diagnostics, supporting earlier and more accurate PD identification.

**a) Handwriting Data**

The Composite Index of Speed and Pen-Pressure (CISP) is a key metric calculated from the data points, offering a robust measure of handwriting quality and motor control. Real-time data capture allows researchers to visualize handwriting and drawing tasks dynamically, replicating the subject's input on the tablet surface. The coordinate and pressure data are stored as CSV files, which can be further analyzed to extract meaningful insights as shown in Figure 8.6.

The feature extraction process is automated, reducing manual effort and ensuring consistency in data analysis. During the recording phase, the system processes coordinate and pressure data packets to compute key metrics. For example, pen-up-down events are tracked to evaluate hand stability, while pressure variations indicate motor control deficits. The extracted features are visualized in real-time using a Python-based graphical interface, which employs Matplotlib for rendering interactive plots. This interface enables researchers to view side-by-side plots of coordinate and pressure data, zoom into specific regions, and export data for further analysis. By providing intuitive tools for data capture and visualization, the system facilitates comprehensive analysis of handwriting impairments associated with PD, as shown in Figure 8.7. The developed system was tested on handwriting data collected from five PD patients and their matched control group (CG). Participants performed spiral drawing tasks using both their affected and unaffected hands, generating extensive datasets for analysis. The results demonstrated significant differences in handwriting features between PD patients and CG

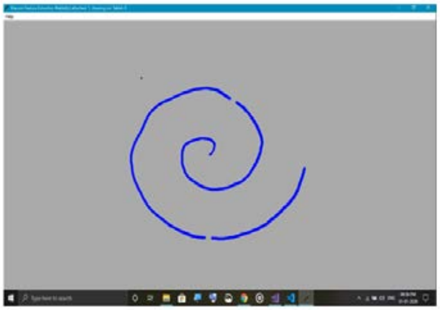

**Fig. 8.7 Pen Ink Visualisation through UI** Pen Ink Visualisation through UI No Permission Required

participants, particularly in pen pressure, speed, and tremor-related metrics. These findings align with established literature, validating the system's effectiveness in capturing PD-related motor impairments. By standardizing spatial alignment and ensuring consistent data collection across participants, the system minimizes variability and enhances the reliability of extracted features.

Generative Adversarial Networks (GANs) are a class of deep learning models in which two neural networks, the Generator and the Discriminator, are trained simultaneously in a minimax game as shown in Figure 8.8. In the context of handwriting generation, GANs were trained to produce realistic synthetic handwriting images that resembled real handwritten text. The objective was to augment existing handwriting datasets by generating realistic synthetic samples, thereby improving the performance of handwriting recognition or verification models by enhancing data diversity, reducing overfitting, and covering rare character styles and variations. The Generator Network (G) took a random noise vector from the latent space or a conditional input such as a text label or style code and produced synthetic handwriting images, learning to generate samples indistinguishable from real handwriting. The Discriminator Network (D) received either a real handwriting image or a fake image from the generator and output a probability indicating whether the image was real (1) or fake (0), learning to distinguish real handwriting from generated samples. Training began with randomly initialized G and D, followed by iterative steps: first training D using real images labelled as real and generated images labelled as fake, then training G to produce handwriting that could fool D. These steps were repeated until the generated samples became indistinguishable from real ones. After training, the generator was used to augment the original dataset with a variety of synthetic handwriting styles, and these enriched datasets were used to train downstream models for handwriting recognition. This approach increased handwriting style diversity (such as slants, thickness, and shapes), improved generalization in recognition systems, and reduced the need for manual data collection.

**b) Feature Extraction from Image Sensor Data**

The video recorded from the CMOS camera is divided into frames to calculate the vibration frequency and processed to obtain the frequency of vibration using the Speeded Up Robust Features (SURF) algorithm. Various image processing steps involved in the estimation of frequency are given in Figure 8.9.

**Surf algorithm to estimate the frequency of vibration**

The working of the SURF algorithm is divided into 4 different stages, and the various steps involved in the calculation of frequency are shown below:

1. **Scale-Space Extrema Detection**: This step identifies local maxima and minima in a scale-space representation, typically denoted as (x, y, γ). Each pixel is compared with its 8 neighboring pixels at the same scale. If a pixel exhibits a maximum or minimum intensity relative to its neighbors, it is classified as an extrema point.

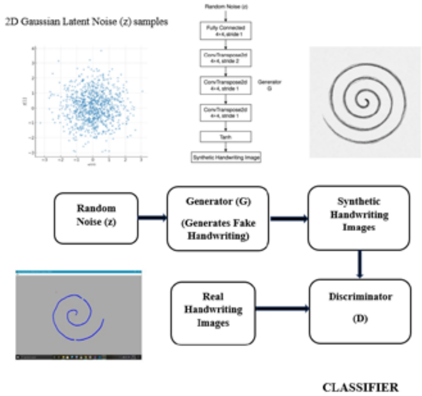


**Fig. 8.8 Implementation of Generative Adversarial Networks for the classification of Handwriting data** Implementation of Generative Adversarial Networks for the classification of Handwriting data No Permission Required

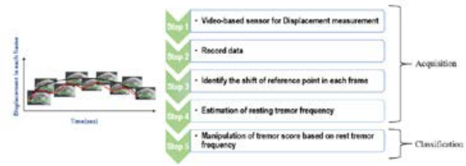


**Fig. 8.9 Steps involved in the calculation of tremor vibration frequency** Steps involved in the calculation of tremor vibration frequency No Permission Required

2. **Keypoint Localization**: This process aims to determine the location and scale of key points in each frame, as illustrated in Figure 8.10. A point is designated as a KeyPoint if it remains stable across multiple frames, meaning its pixel characteristics exhibit minimal variation over time.

**3. Orientation Assignment**: At the identified scale, a histogram of local gradient orientations is constructed to determine the dominant directions. The histogram consists of 36 bins, each representing a 10° interval. Dominant orientations are identified by locating peaks within the histogram. If multiple local peaks exceed 80% of the highest peak, they are also considered, resulting in the generation of multiple keypoints at the same location and scale, each with a distinct orientation.

**4. Keypoint Descriptor**: Each keypoint is represented by its pixel coordinates (x, y). When a keypoint undergoes displacement, the distance between its original and new positions is calculated using the Euclidean distance formula, as expressed in Equation 1

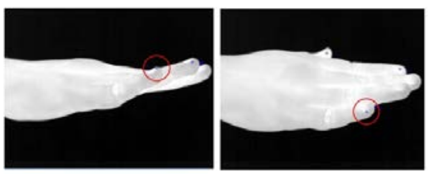

**Fig. 8.10 The marking of key point descriptors in the ROI** The marking of key point descriptors in the ROI No Permission Required

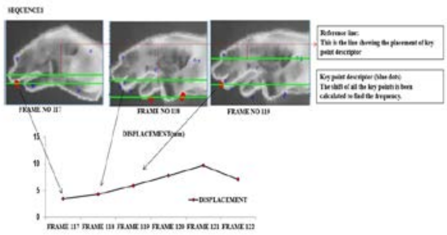

**Fig. 8.11 Consecutive frames, which show the displacement of the key points in hand** Consecutive frames, which show the displacement of the key points in hand No Permission Required

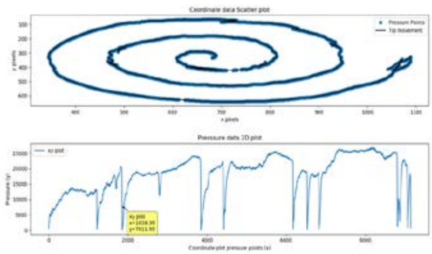


**Fig. 8.12 Analysis of handwriting data** Analysis of handwriting data No Permission Required

$D = \sqrt{(x_2 - x_1)^2 - (y_2 - y_1)^2}(1)$

Based on the location of key point descriptors, as mentioned in Figure 8.11. The displacement of key points can be calculated using Equation 1.

**4. Results and Discussion**

AI-based handwriting analysis has emerged as a promising approach for the early detection and monitoring of Parkinson's Disease (PD), leveraging the ability to identify subtle motor impairments such as tremor, bradykinesia, and micrographia. Many of these systems are developed and trained using publicly available datasets such as the Parkinson's Disease Progression Dataset, which includes a diverse collection of handwriting samples from both healthy individuals and PD patients. These datasets enable machine learning models to recognize distinguishing features associated with disease progression by capturing a wide range of writing characteristics. While these datasets provide a solid foundation for model development, validating AI performance in real-world clinical settings is essential to ensure practical applicability. Clinical trials and patient-specific data are therefore used to assess the system's diagnostic accuracy and reliability in live environments. This validation process is critical not only for confirming the models' effectiveness in detecting PD but also for evaluating their potential for longitudinal disease monitoring. The following sections present the outcomes of handwriting-based assessments and vision-based tremor analyses, along with their integration into AI-driven diagnostic frameworks.

**a) Handwriting Analysis**

The developed handwriting analysis system demonstrated high usability, incorporating an intuitive interface compatible with multiple Wacom devices. Real-time visualizations ( in Figure 8.12) allowed clinicians and researchers to efficiently assess handwriting impairments commonly associated with Parkinson's Disease (PD).

A total of 5 PD subjects and 5 control group (CG) participants were instructed to perform spiral drawing tasks using both their affected and unaffected hands. The system recorded multiple samples, producing over 10,000 KB of raw data. Handwriting inputs were captured using a commercially available Wacom Intuos Pro Large A3-


**Fig. 8.13 CISP Index comparing PWP and CG.** CISP Index comparing PWP and CG. No Permission Required

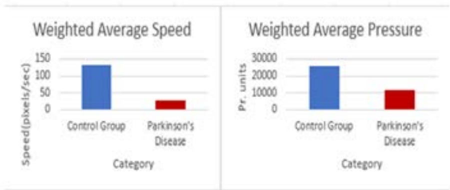


**Fig. 8.14 Average Speed and Average Pressure features for the two groups** Average Speed and Average Pressure features for the two groups No Permission Required

size tablet, with an ink pen capable of detecting (x, y) coordinates and pen pressure (Pr). The lower-left corner of the sheet was designated as the origin and the related observations as shown in Figure 8.13 and Figure 8.14.

Figure 8.15illustrates the training and evaluation results of a GAN-based handwriting classification system. Figure 8.15(b) shows the loss curves for the generator and discriminator over 50 epochs, where both losses decrease and stabilize, indicating effective adversarial training. Figure 8.15(c) chart presents a diversity and style variation analysis, comparing slant, stroke width, curvature, and spacing between original and GAN-generated handwriting, demonstrating that the GAN preserved stylistic diversity close to the original dataset. The Figure 8.15(d) plot displayed recognition accuracy improvement, where a CNN/RNN trained with GAN-augmented data consistently outperformed the baseline without augmentation, achieving an accuracy gain of over 5% by the 10th epoch. The bottom-right plot showed the Equal Error Rate (EER) reduction for handwriting verification, with GAN augmentation lowering the EER steadily across epochs, suggesting fewer false acceptances and rejections. Overall, the results indicated that GAN augmentation enhanced recognition performance, preserved handwriting style diversity, and improved verification robustness.

The model's predictions were cross-validated against Unified Parkinson's Disease Rating Scale (UPDRS) scores, revealing a strong correlation between AI-detected impairments and clinical severity ratings. Additionally, the deployment of this system on mobile and wearable platforms enabled real-time monitoring, offering a low-cost, scalable solution for continuous tracking of motor symptom progression. These advancements supported the development of an accurate, data-driven framework for early detection and ongoing management of Parkinson's disease, bridging the gap between subjective clinical evaluation and objective digital biomarkers.

**b) Tremor analysis using a vision-based sensor**

Video-based tremor analysis was performed using Gaussian filtering to reduce high-frequency noise, followed by Gaussian pyramids for multi-scale motion representation. Frame-by-frame displacement was tracked to estimate tremor periodicity. The average recorded tremor frequency was 8.2 Hz under experimental conditions ( Table 8.2).

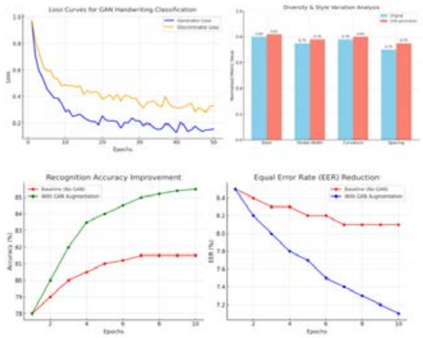


**Fig. 8.15 Performance Metrics (a) Loss Curves for GAN Handwriting Classification(b) Diversity & Style Variation Analysis(c) Recognition Accuracy Improvement(d) Equal Error Rate (EER) Reduction** Analysis No Permission Required

**Table 8.2** Calculation of resting tremor frequency using the CMOS sensor

| Frame number | Time Period(s) | Displacement (mm) | Acceleration (mm/sq. Sec) | Frequency:(Hz) |
|---|---|---|---|---|
| 117 | 0.04 | 3.5 | 66.1 | 5.2 |
| 118 | 0.08 | 4.3 | 73.3 | 9.2 |
| 119 | 0.12 | 5.9 | 85.8 | 8.5 |
| 120 | 0.16 | 7.8 | 98.7 | 8.0 |
| 121 | 0.2 | 9.7 | 110.1 | 7.5 |
| 122 | 0.24 | 7.1 | 94.2 | 8.1 |

*No Permission Required*

**Table 8.3** Calculation of resting tremor frequency using the thermal sensor

| Frame number | Time Period(s) | Displacement (mm) | Acceleration (mm/sq. Sec) | Frequency (Hz) |
|---|---|---|---|---|
| 611 | 0.04 | 7 | 43.75 | 5.6 |
| 612 | 0.08 | 14 | 55 | 4.4 |
| 613 | 0.12 | 18 | 73.7 | 2.2 |
| 614 | 0.16 | 29 | 108.75 | 4.3 |
| 615 | 0.2 | 32 | 306 | 6.9 |
| 616 | 0.24 | 34 | 123 | 4.2 |

*No Permission Required*

**Table 8.4** The tremor frequency observed in 15 subjects

| PATIENT ID | DISPLACEMENT (Mean ± SD) (m) | FREQUENCY (Mean ± SD) (Hz) | POSITION OF TREMOR |
|---|---|---|---|
| 1 | 9856.34±0.15 | 5.181±0.034 | Hands on lap |
| 2 | 17438.49±0.25 | 3.556±0.355 | While lying down |
| 3 | 438.49±0.42 | 21.967±0.06 | Hands stretched out |
| 4 | 658.9±1.32 | 24.955±0.144 | Hands streched out |
| 5 | 9984.12±0.84 | 7.144±1.320 | Hands on lap |
| 6 | 4589.12±0.19 | 12.539±0.089 | Hands stretched out |
| 7 | 8253.83±0.36 | 10.673±0.565 | While lying down |
| 8 | 10673.98±0.73 | 20.947±0.211 | While lying down |
| 9 | 35938.98±0.17 | 8.106±0.003 | Hands on lap |
| 10 | 84735.2±0.49 | 3.800±0.078 | Hands stretched out |
| 11 | 9005.02±2.09 | 5.745±0.885 | While lying down |
| 12 | 12647.3±0.41 | 13.640±1.880 | Hands stretched out |
| 13 | 11946.31±0.86 | 7.703±1.515 | Hands on lap |
| 14 | 9976.21±0.41 | 4.935±0.264 | While lying down |
| 15 | 46288.45±0.81 | 11.566±1.221 | Hands on lap |

*No Permission Required*

Data were captured using both standard CMOS and thermal infrared sensors. For an input vibration at 4 Hz, the CMOS system estimated a frequency of ~4.3 Hz, whereas the thermal sensor exhibited lower error margins and an improved signal-to-noise ratio (Table 8.3). These findings indicate thermal imaging as a superior method due to its ability to suppress visual clutter and detect subcutaneous muscle activity.

In Table 8.4, The data presents measured tremor characteristics for different individuals, focusing on two primary parameters: displacement (amplitude of the tremor in millimeters) and frequency (oscillation rate in hertz). These measurements are used to differentiate between three tremor types: essential tremor (ET), typically in the 4–12 Hz range with moderate displacement; Parkinson's tremor (PT), usually between 4–6 Hz with lower displacement; and physiological tremor (PHT), occurring at higher frequencies (7–30 Hz) with relatively small amplitude. Each row in the dataset corresponds to an observation, containing numerical values for displacement and frequency, along with a class label identifying the tremor type. This dataset serves as the foundation for training the deep learning classification model, where the numerical features guide the network in learning patterns that distinguish the three categories. The tremor classification system begins with data input, where patient tremor measurements, specifically displacement and frequency, are collected to distinguish between essential tremor (4–12 Hz), Parkinson's tremor (4–6 Hz), and physiological tremor (7–30 Hz). The preprocessing stage involves cleaning the data, extracting features (with frequency as the primary and displacement as a secondary input), normalizing values, and encoding class labels numerically. A deep learning classifier, implemented in either TensorFlow with Keras for rapid prototyping or PyTorch for research flexibility, processes the input through an architecture with two input neurons (displacement and frequency), multiple hidden layers using ReLU activation, and a softmax output layer producing probabilities for each class. The model is trained using categorical cross-entropy loss and the Adam optimizer, with performance measured through accuracy and other metrics. After training on a split of the dataset, the system outputs class probabilities for each patient, enabling precise prediction of tremor type.

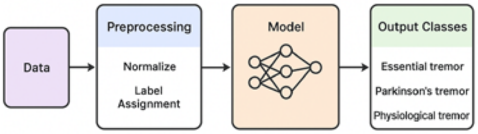


**Fig. 8.16 Tremor classification using deep learning model**
Tremor classification using deep learning model No Permission Required

The tremor classification model can be effectively implemented using TensorFlow with Keras by designing a simple yet efficient neural network architecture, as shown in Figure 8.16. The input layer consists of two neurons corresponding to the measured displacement and frequency values from the dataset. This is followed by two to three dense hidden layers with ReLU activation functions, enabling the network to capture nonlinear relationships between features. The output layer has three neurons, each representing one tremor type—essential tremor, Parkinson's tremor, and physiological tremor—and uses a softmax activation to produce class probability distributions. The model is trained using the categorical cross-entropy loss function, with the Adam optimizer for adaptive learning rate adjustments, and performance is monitored through metrics such as accuracy, precision, and recall. The dataset is split into 80% training and 20% testing sets, and training is conducted over 50–100 epochs depending on convergence behavior, with a small batch size (e.g., 4) due to the limited dataset size. To improve generalization and mitigate overfitting, data augmentation is applied by generating synthetic variations of the existing measurements using statistical perturbations. This approach leverages both frequency and displacement for improved accuracy, supports flexibility in development environments, and is scalable to larger datasets or additional biomedical features. Features such as tremor frequency, amplitude, posture symmetry, and tremor positioning were extracted and used to train machine learning and deep learning models, including K-Nearest Neighbors (KNN), Decision Trees, and neural network architectures. The dataset included healthy controls, PD patients, and individuals with other neurological disorders. Model outputs included disease stage classification and generation of MDS-UPDRS scores (0–4 scale). The system achieved high diagnostic performance, with an AUC of 0.987, a Precision of 0.927, and an F1-score of 0.909. Performance metrics were evaluated using standard measures including sensitivity, specificity, and ROC analysis (Figure 8.17.). Cross-validation with expert neurologist assessments confirmed the clinical relevance and reliability of the system.

The classification results show that the model achieves moderate overall performance, with a macro-average precision of 0.97, indicating that when averaging across all classes equally, the model's predictions are only correct 40% of the time. The macro-average recall of 0.97 suggests that it successfully identifies half of the actual cases across all disorder types, while the macro-average F1-score of 0.96 reflects a balance between these two metrics. Improving the dataset balance and applying targeted data augmentation could help enhance the model's generalization and performance across all categories. The results from both handwriting and vision-based tremor assessments highlight the potential of AI-integrated, non-invasive diagnostic frameworks for early-stage PD detection. The handwriting system provides a fine-grained, scalable assessment of upper limb motor impairments, while vision-based tremor analysis offers a contactless solution with good accuracy. The combined use of these modalities, validated against clinical benchmarks such as the UPDRS, offers a robust diagnostic toolset. These findings support the broader application of AI-enabled systems in clinical and remote

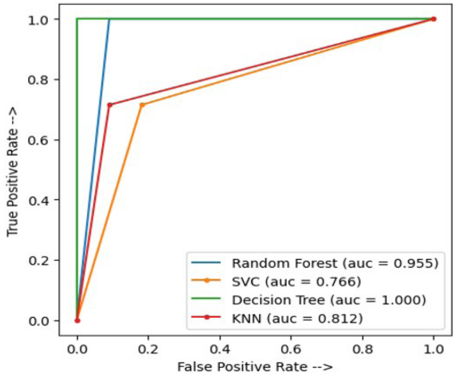


**Fig. 8.17 Performance estimation of the classifiers**
Performance estimation of the classifiers No Permission Required

monitoring environments. Future work will focus on expanding the dataset across demographics, incorporating additional physiological markers (e.g., EMG, HRV), and leveraging advanced architectures such as CNN-LSTM hybrids to support real-time, longitudinal disease monitoring.

**5. Conclusion**

PD is a progressive neurodegenerative disorder marked by the degeneration of dopaminergic neurons. It presents significant challenges in early diagnosis due to its subtle symptom onset and the absence of standardized, objective diagnostic tools. While clinical assessments such as the UPDRS remain the benchmark for evaluating PD severity, they rely heavily on subjective patient responses and clinical judgment, often leading to diagnostic delays and elevated healthcare costs. In recent years, numerous studies have explored both imaging and non-imaging techniques, including thermal imaging, 2D/3D camera-based gait analysis, and wearable IMUs to identify motor abnormalities associated with early PD, particularly resting tremor, bradykinesia, and gait irregularities. However, many of these approaches lack clinical validation and often function as standalone diagnostic tools without integration into a unified, multimodal framework. Furthermore, very few studies benchmark their results against the UPDRS, thus limiting the clinical relevance and translational impact of their findings. This study presents a comprehensive, AI-driven diagnostic framework to evaluate early PD symptoms. A key innovation lies in the validation of all assessment modules against UPDRS scoring, ensuring alignment with established clinical standards. By incorporating deep learning algorithms, the system enables automated feature extraction and classification from multiple sensor sources, supporting reliable and objective detection of PD-related motor impairments. Experimental results demonstrated that the proposed system achieved high accuracy, sensitivity, and specificity in distinguishing PD subjects from healthy controls. The machine-generated symptom scores showed strong correlation with neurologist-rated UPDRS scores, validating the effectiveness of the multimodal diagnostic model. This integrated approach not only outperforms conventional single-modality techniques but also offers significant potential for real-world clinical deployment.

Despite its considerable potential, AI-based handwriting analysis faced several challenges, particularly in uncontrolled, real-world environments. These included sensor noise, interpersonal variability, and inconsistencies in data quality. Similarly, the implementation of machine learning models in clinical settings often suffers from overfitting, limited generalizability, and regulatory constraints. These limitations underscore the need for ongoing model refinement, external validation across diverse and demographically representative populations, and strict compliance with established clinical standards. To address these challenges, research efforts focused on expanding datasets to include broader population segments and all stages of Parkinson's disease (PD). Emphasis was placed on developing portable, real-time diagnostic tools that integrated with wearable sensors and mobile devices, enabling continuous, home-based monitoring of symptom progression. Adaptive learning approaches were explored to ensure consistent performance despite fluctuations in patient behavior or sensor conditions. To overcome data scarcity, particularly for rare or early-stage symptoms, Generative Adversarial Networks (GANs) were used to generate realistic synthetic data to augment the training datasets and improve model generalizability. Collectively, these advancements contributed to the development of a robust, objective diagnostic framework that complemented traditional clinical evaluations. The application of GANs in handwriting analysis proved to be an effective strategy for enhancing both recognition and verification systems. By generating realistic and diverse handwriting samples, the GAN mitigated data scarcity and enriched the training set with varied styles, leading to notable gains in recognition accuracy and reductions in equal error rates. The style diversity assessment confirmed that the GAN avoided significant mode collapse and maintained close alignment with the original handwriting characteristics. These results indicated that GAN-based augmentation not only improved model generalization and robustness but also preserved the authenticity of handwriting patterns, making it a valuable tool for advancing handwriting analysis tasks. By leveraging multimodal sensor data and state-of-the-art AI techniques, the system enabled earlier detection, more timely interventions, and personalized treatment plans for individuals with Parkinson's disease, ultimately supporting improved patient outcomes and quality of life.

**AI Disclosure**

Artificial intelligence (AI) tools, including OpenAI's ChatGPT, were utilized to assist in the preparation of this manuscript. These tools supported tasks such as language enhancement, grammatical refinement, and content organization. All intellectual contributions, data analysis, and final decisions were made independently by the authors, who assume full responsibility for the accuracy and integrity of the work presented.

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